

Student Takeaway for Sync Session

<<Week 7: Programming Fundamentals for Machine Learning - Python>>

This guide is your roadmap to making the most of our online session. Packed with essential tips and strategies, it's designed to keep you engaged, prepared, and ready to dive into a smooth and productive learning journey. Get ready to participate, learn, and thrive!

Session Overview

Session title	Programming Fundamentals for Machine Learning - Python		
Session duration	3 hours		
Session type	• Lectures: Conceptual understanding of NumPy, Pandas, and Matplotlib for data handling and visualisation. • Exercises: Practical implementation through coding examples and exercises.		
Scope	This session introduces the foundational Python libraries used in machine learning, including: • Core operations with NumPy arrays • Data manipulation using Pandas • Data visualisation using Matplotlib and Seaborn • Hands-on exercises for basic EDA and visualisations		
Learning objectives	Objective	Core capability	
	Understand and apply NumPy arrays for efficient computation	Numerical thinking and array operations	
	Manipulate datasets using Pandas	Data wrangling and cleaning	
	Perform vectorised operations using NumPy and Pandas	Efficient coding and data handling	
	Visualise data using Matplotlib and Seaborn	Communicating data insights visually	
	Read and process CSV, Excel, JSON, SQL files	Data ingestion techniques	
Software/Tools	• Explore the Data: Understand the dataset before coding. • Practice & Modify: Replicate examples, then tweak them. • Use Docs & Debug: Refer to documentation and debug smartly. • Keep It Clean: Write clear, well-commented code.		

Pro tips for success:

- **Ask Bold Questions:** No question is too small—curiosity drives learning!
- **Be Hands-On:** Experiment with code, tweak parameters, and observe changes.
- **Collaborate:** Share ideas in discussions—your perspective can spark insights!

Session Details

Slide(s)	Topic/Issue	What to expect	Notes
1	Introduction	Overview of session objectives	What are programming fundamentals for ML?
2-3	NumPy Overview	Why NumPy? Efficiency and utility in ML	Fast array operations
4-5	NumPy Arrays	ndarray and storage features	Multi-dimensional array examples
6-7	Indexing and Slicing	Access elements and subarrays	1D and 2D example
8-9	Arithmetic Ops	Element-wise operations	Add, subtract, multiply arrays
10-11	Broadcasting	Working with different shapes	Scalar and array example
12-13	Universal Functions	Trig, exponential, and logarithmic functions	sin, log, log10
14-15	Array Manipulation	Transpose, concatenate, reshape	Efficient array handling
16-17	Matrix Multiplication	Using dot() and @	Linear algebra in NumPy
18	Matrix Inversion	Using linalg.inv	Validation with identity matrix
19	Matrix Decomposition	Eigenvalues and eigenvectors	Reconstruct matrix
20	NumPy in Action	Applications in ML and data analysis	Real-world examples
21	Pandas Overview	Features and advantages	Why use Pandas?
22	Series Structure	1D labelled array	Labels and data types
23	Creating Series	List, dictionary, custom index	Init with examples
24	Accessing Elements	Label and position access	Slicing
25	Vectorised Ops	Arithmetic and math ops	NumPy integration

Slide(s)	Topic/Issue	What to expect	Notes
26	Series Methods	sum(), mean(), value_counts(), apply()	Aggregation and custom logic
27	Missing Data	isna(), fillna(), dropna()	Real-life data fixing
28	DataFrames	2D structure, columns with types	Dictionary of Series
29	Reading CSV	pd.read_csv() and .head()	Importing data
30	Reading Excel	pd.read_excel() with sheet	File reading
31	Reading JSON	pd.read_json()	Auto-tabular format
32	Reading SQL	Connect, query, and load	Use sqlite3
33	Reading Tips	na_values, usecols, chunksize	Custom loading
34	Error Handling	try-except for robustness	FileNotFoundError example
35	DataFrame Ops	View, filter, add/drop cols	Basic operations
36	Exercise 1	Load CSV, use info() and describe()	sales_data.csv
37	Exercise 2	Accessing data, filtering rows	head(), unique(), filter
38	Exercise 3	Filter by region, price, sales	Conditionals
39	Exercise 4	Missing value handling	Median replacement and dropna()
40	Exercise 5	Add Discounted_Price column	10% reduction
41	Exercise 6	Drop column	Verify with columns
42	Exercise 7	NumPy functions on data	Profit and log

Post session

Reflection challenge	What did you enjoy the most about hands-on programming in Python?
Explore more	• Read: Python Cookbook sections on data analysis and visualisation. • Watch: Real-world ML application tutorials using Pandas and Matplotlib. • Play around: Use a Kaggle dataset to practice what you've learned.
Get inspired	Did you know Seaborn and Matplotlib help visualise insights that drive decisions in climate research, sports analytics, and healthcare? These skills empower you to make data come alive.
The journey ahead	In the upcoming session, we delve into Probability and Statistics—enabling you to handle uncertainty, build smarter models, and gain deeper insights from data. Next, dive into Essential Concepts in Probability and Statistics, where you'll learn about probability distributions, statistical inference, and their role in ML algorithms.

For learner use only